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Nootropic Drugs

Stuart Silverman

A comprehensive review for information on nootropic drugs. Search covered Medline, EMBASE and Biosis and looked at those drugs that are classified as nootropic or smart drugs, memory enhancers or cognitive enhancers. Linking to full text or the document ordering system (Get@Pfizer) are embedded in the title of each record.

Table of Contents

1. The pharmacoeconomics of cognitive enhancers in moderate to severe Alzheimer's disease 5

2. Fluoxetine potentiates methylphenidate-induced gene regulation in addiction-related brain regions: concerns for use of cognitive enhancers? 5

3. Piracetam and piracetam-like drugs: from basic science to novel clinical applications to CNS disorders 6

4. Smart drugs for cognitive enhancement: ethical and pragmatic considerations in the era of cosmetic neurology.[Erratum appears in J Med Ethics. 2009 Dec;35(12):738] 7

5. Can cognitive enhancers reduce the risk of falls in older people with mild cognitive impairment? A protocol for a randomised controlled double blind trial 7

6. The inhibitory avoidance test optimized for discovery of cognitive enhancers 8

7. Towards an animal model of an antipsychotic drug-resistant cognitive impairment in schizophrenia: scopolamine induces abnormally persistent latent inhibition, which can be reversed by cognitive enhancers but not by antipsychotic drugs 9

8. Bioequivalence and pharmacokinetic comparison of a single 200-mg dose of meclofenoxate hydrochloride capsule and tablet formulations in healthy Chinese adult male volunteers: a randomized sequence, open-label, two-period crossover study 9

9. Strain-dependent effects of cognitive enhancers in the mouse 10

10. The use of cognitive enhancers in behavioral disturbances of Alzheimer's disease 11

11. Piracetam--an old drug with novel properties? 11

12. Inefficacy of piracetam in the prevention of painful crises in children and adolescents with sickle cell disease 12

13. Piracetam and TRH analogues antagonise inhibition by barbiturates, diazepam, melatonin and galanin of human erythrocyte D-glucose transport 12

14. A pilot trial of piracetam and ginkgo biloba for the treatment of cocaine dependence 13

15. Smart drugs: tyrosine kinase inhibitors in cancer therapy 13

16. Piracetam for dementia or cognitive impairment 14

17. Piracetam affects facilitatory I-wave interaction in the human motor cortex 15

18. Piracetam and levetiracetam: close structural similarities but different pharmacological and clinical profiles 15

19. Piracetam in the treatment of acute stroke 16

20. [The 1998 domestic state of development of cognitive enhancers] 16

21. Piracetam in the treatment of schizophrenia: implications for the glutamate hypothesis of schizophrenia 17

22. Single-dose piracetam effects on global complexity measures of human spontaneous multichannel EEG .. 17

23. ['Smart drugs' enticements on the Internet] 18

24. Functional activation of cerebral blood flow abolished by scopolamine is reversed by cognitive enhancers associated with cholinesterase inhibition: a positron emission tomography study in unanesthetized monkeys 18

25. [A critical review of available tools for evaluating memory enhancers in Alzheimer's disease] 18

26. Alterations in acetylcholine, NMDA, benzodiazepine receptors and protein kinase C in the brain of the senescence-accelerated mouse: an animal model useful for studies on cognitive enhancers 19

27. Cognitive enhancers and hippocampal long-term potentiation in vitro 19

28. Cognitive enhancers in theory and practice: studies of the cholinergic hypothesis of cognitive deficits in Alzheimer's disease 20

29. Nootropic agents stimulate neurogenesis 20

30. Epibatidine - Design for new cognitive enhancers. [Czech] Epibatidin - Model pro nova kognitiva 21

31. Surfing with side effects. Problems with smart drugs. [Dutch] Surfen Met Bijwerkingen. Problemen Rond Smart Drugs 21

32. Piracetam: A review of its clinical potential in the management of patients with stroke 22

33. Piracetam in the treatment of acute stroke 22

34. Pharmacological properties of piracetam: Rationale for use in stroke patients 23

35. 'Smart drugs': Misleading advertising on internet. [Dutch] 'Smart drugs': De verlokkingen van het internet 24

36. Treatment of acute ischemic stroke with piracetam 24

37. A critical review of available measurement tools for assessing memory enhancers in patients with Alzheimer's disease. [French] Une revue critique des instruments disponibles pour l'evaluation des medicaments promnesiants dans la maladie d'Alzheimer 25

38. Piracetam in patients with chronic vertigo. Results of a double-blind, placebo-controlled study 25

39. Cortical myoclonus: A valid human model for the demonstration of the neuromodulatory effects of piracetam 26

40. Piracetam in developmental reading disorders: A review 26

41. Effect of piracetam on recovery and rehabilitation after stroke: A double- blind, placebo-controlled study 27

42. Superiority of Antagonic-Stress composition versus nicergoline in gerontopsychiatry 27

43. Smart Drugs: Implications of student use 28

44. Effects of oxiracetam on organic brain syndrome following activation of the high-affinity choline uptake system: A neurochemical analysis of the cerebrospinal fluid of four patients 28

45. Long-term and high-dose piracetam treatment of Alzheimer's disease 29

46. Effect of CDP-choline on learning and memory processes in rodents 29

47. Clinical use of piracetam in epileptic patients 30

48. Effects of four non-cholinergic cognitive enhancers in comparison with tacrine and galanthamine on scopolamine-induced amnesia in rats 30

49. Piracetam. An overview of its pharmacological properties and a review of its therapeutic use in senile cognitive disorders 31

50. Clinical issues of cognitive enhancers in Alzheimer disease 31

51. The use of a scopolamine model to study the potential nootropic effects of aniracetam and piracetam in healthy volunteers 32

52. A controlled trial of piracetam in intellectually impaired patients with Parkinson's disease 32

53. Choosing a target for cognitive enhancers 32

54. An hypothesis on the role of glucose in the mechanism of action of cognitive enhancers 33

55. Piracetam: physiological disposition and mechanism of action 33

56. Piracetam combined with lecithin in the treatment of Alzheimer's disease 34

57. The red cell as a model for studying neurotropic drugs: Effects of piracetam on erythrocyte membrane 34

58. The effects of oxiracetam (ISF 2522) in patients with organic brain syndrome. (A double-blind controlled study with piracetam) 34

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1.The pharmacoeconomics of cognitive enhancers in moderate to severe Alzheimer's disease

CNS Drugs. 24 (11) pp. 909-27; 2010.

Jaclyn Cappell, Nathan Herrmann, Stephen Cornish and Krista L. Lanctot
Department of Psychiatry, Sunnybrook Health Sciences Centre, Toronto, Ontario, Canada.

Abstract
Alzheimer's disease is associated with a substantial economic impact on patients, their caregivers and society. Due to the current rise in the aging population, the prevalence and impact of Alzheimer's disease are expected to increase greatly. The cost of caring for someone with Alzheimer's disease is magnified in the more severe stages of the disease. There are four cognitive enhancers commonly used for the treatment of Alzheimer's disease: three cholinesterase inhibitors (donepezil, rivastigmine and galantamine) and one NMDA receptor antagonist (memantine). Of these, donepezil and memantine have been approved in many countries as pharmacological treatments for moderate to severe Alzheimer's disease, while donepezil, rivastigmine and galantamine are approved treatments for mild to moderate Alzheimer's disease. While cost effectiveness has been well studied in mild to moderate Alzheimer's disease, the cost-benefit information for drug therapy in moderate to severe Alzheimer's disease is less clear. This article reviews the pharmacoeconomic data available on these four drugs, with a specific focus on moderate to severe Alzheimer's disease, including economic burden, cost drivers, clinical outcomes and pharmacoeconomic studies. A key driver of the cost of Alzheimer's disease is the severity of the disease, indicating that the ability to stabilize the disease state is a potential source of cost savings. Drug therapies that can limit increases in behavioural problems and cognitive and functional impairment, and postpone institutionalization without an increase in longevity may serve to reduce the economic burden on Alzheimer's disease patients. The data suggest that, while the available, approved agents offer only modest improvements in clinical outcomes, they could be cost-effective treatments for moderate to severe Alzheimer's disease when viewed from the societal perspective. For memantine and donepezil, data are available that suggest that the cost of these drugs is offset by the clinical and societal benefits provided by slowing the progression of Alzheimer's disease. While there are few head-to-head comparison trials, the similarity in costs of the treatments and efficacy against placebo suggest that cost effectiveness will not be substantially different among treatments. More studies that examine longitudinal resource utilization and its relationship to drug treatment in the moderate to severe stages are needed to clarify cost benefit in this population and possibly differentiate between individual medications.

2.Fluoxetine potentiates methylphenidate-induced gene regulation in addiction-related brain regions: concerns for use of cognitive enhancers?

Biological Psychiatry. 67 (6) pp. 592-4; 2010.

Heinz Steiner, Vincent Van Waes and Michela Marinelli
Department of Cellular and Molecular Pharmacology, Rosalind Franklin University of Medicine and Science, Chicago Medical School, North Chicago, Illinois, USA. Heinz.Steiner@rosalindfranklin.edu

Abstract
BACKGROUND: There is growing use of psychostimulant cognitive enhancers such as methylphenidate

(Ritalin). Methylphenidate differs from the psychostimulant cocaine because it does not enhance synaptic levels of serotonin. We investigated whether exposure to methylphenidate combined with a serotonin-enhancing medication, the prototypical selective serotonin reuptake inhibitor (SSRI) fluoxetine (Prozac), would produce more "cocaine-like" molecular and behavioral changes. METHODS: We measured the effects of fluoxetine on gene expression induced by the cognitive enhancer methylphenidate in the striatum and nucleus accumbens of rats, by in situ hybridization histochemistry. We also determined whether fluoxetine modified behavioral effects of methylphenidate. RESULTS: Fluoxetine robustly potentiated methylphenidate-induced expression of the transcription factors c-fos and zif 268 throughout the striatum and to some degree in the nucleus accumbens. Fluoxetine also enhanced methylphenidate-induced stereotypical behavior. CONCLUSIONS: Both potentiated gene regulation in the striatum and the behavioral effects indicate that combining the SSRI fluoxetine with the cognitive enhancer methylphenidate mimics cocaine effects, consistent with an increased risk for substance use disorder. Copyright 2010 Society of Biological Psychiatry. Published by Elsevier Inc. All rights reserved.

3.Piracetam and piracetam-like drugs: from basic science to novel clinical applications to CNS disorders

Drugs. 70 (3) pp. 287-312; 2010.

Andrei G. Malykh and M. Reza Sadaie
NovoMed Consulting, Silver Spring, Maryland 20904, USA.

Abstract

There is an increasing interest in nootropic drugs for the treatment of CNS disorders. Since the last meta-analysis of the clinical efficacy of piracetam, more information has accumulated. The primary objective of this systematic survey is to evaluate the clinical outcomes as well as the scientific literature relating to the pharmacology, pharmacokinetics/pharmacodynamics, mechanism of action, dosing, toxicology and adverse effects of marketed and investigational drugs. The major focus of the literature search was on articles demonstrating evidence-based clinical investigations during the past 10 years for the following therapeutic categories of CNS disorders: (i) cognition/memory; (ii) epilepsy and seizure; (iii) neurodegenerative diseases; (iv) stroke/ischaemia; and (v) stress and anxiety. In this article, piracetam-like compounds are divided into three subgroups based on their chemical structures, known efficacy and intended clinical uses. Subgroup 1 drugs include piracetam, oxiracetam, aniracetam, pramiracetam and phenylpiracetam, which have been used in humans and some of which are available as dietary supplements. Of these, oxiracetam and aniracetam are no longer in clinical use. Pramiracetam reportedly improved cognitive deficits associated with traumatic brain injuries. Although piracetam exhibited no long-term benefits for the treatment of mild cognitive impairments, recent studies demonstrated its neuroprotective effect when used during coronary bypass surgery. It was also effective in the treatment of cognitive disorders of cerebrovascular and traumatic origins; however, its overall effect on lowering depression and anxiety was higher than improving memory. As add-on therapy, it appears to benefit individuals with myoclonus epilepsy and tardive dyskinesia. Phenylpiracetam is more potent than piracetam and is used for a wider range of indications. In combination with a vasodilator drug, piracetam appeared to have an additive beneficial effect on various cognitive disabilities. Subgroup 2 drugs include levetiracetam, seletracetam and brivaracetam, which demonstrate antiepileptic activity, although their cognitive effects are unclear. Subgroup 3 includes piracetam derivatives with unknown clinical efficacies, and

of these nefiracetam failed to improve cognition in post-stroke patients and rolipram is currently in clinical trials as an antidepressant. The remaining compounds of this subgroup are at various preclinical stages of research. The modes of action of piracetam and most of its derivatives remain an enigma. Differential effects on subtypes of glutamate receptors, but not the GABAergic actions, have been implicated. Piracetam seems to activate calcium influx into neuronal cells; however, this function is questionable in the light of findings that a persistent calcium inflow may have deleterious impact on neuronal cells. Although subgroup 2 compounds act via binding to another neuronal receptor (synaptic vesicle 2A), some of the subgroup 3 compounds, such as nefiracetam, are similar to those of subgroup 1. Based on calculations of the efficacy rates, our assessments indicate notable improvements in clinical outcomes with some of these agents. [References: 151]

4.Smart drugs for cognitive enhancement: ethical and pragmatic considerations in the era of cosmetic neurology.[Erratum appears in J Med Ethics. 2009 Dec;35(12):738]

Journal of Medical Ethics. 35 (10) pp. 611-5; 2009.

V. Cakic
University of Sydney, Australia. vincec@psych.usyd.edu.au

Abstract

Reports in the popular press suggest that smart drugs or "nootropics" such as methylphenidate, modafinil and piracetam are increasingly being used by the healthy to augment cognitive ability. Although current nootropics offer only modest improvements in cognitive performance, it appears likely that more effective compounds will be developed in the future and that their off-label use will increase. One sphere in which the use of these drugs may be commonplace is by healthy students within academia. This article reviews the ethical and pragmatic implications of nootropic use in academia by drawing parallels with issues relevant to the drugs in sport debate. It is often argued that performance-enhancing drugs should be prohibited because they create an uneven playing field. However, this appears dubious given that "unfair" advantages are already ubiquitous and generally tolerated by society. There are concerns that widespread use will indirectly coerce non-users also to employ nootropics in order to remain competitive. However, to restrict the autonomy of all people for fear that it may influence the actions of some is untenable. The use of potentially harmful drugs for the purposes of enhancement rather than treatment is often seen as unjustified, and libertarian approaches generally champion the rights of the individual in deciding if these risks are acceptable. Finally, whether the prohibition of nootropics can be effectively enforced is doubtful. As nootropics use becomes widespread among students in the future, discussion of this issue will become more pressing in the years to come.

5.Can cognitive enhancers reduce the risk of falls in older people with mild cognitive impairment? A protocol for a randomised controlled double blind trial

BMC Neurology. 9 pp. 42; 2009.

Manuel Montero-Odasso, Jennie L. Wells, Michael J. Borrie and Mark Speechley
Department of Medicine, Division of Geriatric Medicine, Parkwood Hospital, University of Western Ontario, London, ON, Canada. Manuel.MonteroOdasso@sjhc.london.on.ca

Abstract

BACKGROUND: Older adults with cognitive problems have a higher risk of falls, at least twice that of

cognitively normal older adults. The consequences of falls in this population are very serious: fallers with cognitive problems suffer more injuries due to falls and are approximately five times more likely to be admitted to institutional care. Although the mechanisms of increased fall risk in cognitively impaired people are not completely understood, it is known that impaired cognitive abilities can reduce attentional resource allocation while walking. Since cognitive enhancers, such as cholinesterase inhibitors, improve attention and executive function, we hypothesise that cognitive enhancers may reduce fall risk in elderly people in the early stages of cognitive decline by improving their gait and balance performance due to an enhancement in attention and executive function. METHOD/DESIGN: Double blinded randomized controlled trial with 6 months follow-up in 140 older individuals with Mild Cognitive Impairment (MCI). Participants will be randomized to the intervention group, receiving donepezil, and to the control group, receiving placebo. A block randomization by four and stratification based on fall history will be performed. Primary outcomes are improvements in gait velocity and reduction in gait variability. Secondary outcomes are changes in the balance confidence, balance sway, attention, executive function, and number of falls. DISCUSSION: By characterizing and understanding the effects of cognitive enhancers on fall risk in older adults with cognitive impairments, we will be able to pave the way for a new approach to fall prevention in this population. This RCT study will provide, for the first time, information regarding the effect of a medication designed to augment cognitive functioning have on the risk of falls in older adults with Mild Cognitive Impairment. We expect a significant reduction in the risk of falls in this vulnerable population as a function of the reduced gait variability achieved by treatment with cognitive enhancers. This study may contribute to a new approach to prevent and treat fall risk in seniors in early stages of dementia. TRIAL REGISTRATION: The protocol for this study is registered with the Clinical Trials Registry, identifier number: NCT00934531 http://www.clinicaltrials.gov.

6.The inhibitory avoidance test optimized for discovery of cognitive enhancers

Behavior Research Methods. 41 (3) pp. 805-11; 2009.

Eric R. A. Y. Detrait, Etienne Hanon, Bertrand Dardenne and Yves Lamberty
CNS Pharmacology Department-Cognitive Psychopharmacology, UCB Pharma, Braine-l'Alleud, Belgium.
eric.detrait@ucb-group.com

Abstract
In the present article, we describe a new protocol for the inhibitory avoidance test, with a dual purpose: (1) to provide a less variable and more reliable assessment of the efficacy of potential cognitive enhancers in antagonizing scopolamine-induced long-term-memory deficits, and (2) to secure a high throughput for pharmacological screening of cognitive enhancers. The new protocol consists of two acquisition trials that are followed 24 h later by a single retention trial. In the present study, this protocol clearly dissociated the frequency distributions of retention latencies between scopolamine- and vehicle-treated groups and allowed validation by means of two acetylcholinesterase inhibitors-tacrine and donepezil-that proved to be active in counteracting the scopolamine-induced memory deficit. This protocol also produced stability of the behavioral response to pharmacological agents over a 3-year period. A statistical power analysis indicated that, depending on the efficacy of the drug/dose, a sample size of 5-12 mice was required in order to show a reversal of the scopolamine-induced memory deficit. The double-trial acquisition protocol is suitable for testing cognitive enhancers, while also providing a clearly enhanced throughput.

7.Towards an animal model of an antipsychotic drug-resistant cognitive impairment in schizophrenia: scopolamine induces abnormally persistent latent inhibition, which can be reversed by cognitive enhancers but not by antipsychotic drugs

International Journal of Neuropsychopharmacology. 12 (2) pp. 227-41; 2009.

Segev Barak and Ina Weiner
Department of Psychology, Tel-Aviv University, Tel-Aviv, Israel.

Abstract
Schizophrenia symptoms segregate into positive, negative and cognitive, which exhibit differential sensitivity to drugs. Recent efforts to identify treatments targeting cognitive impairments in schizophrenia have directed attention to the cholinergic system for its well documented role in cognition. Relatedly, muscarinic antagonists (e.g. scopolamine) produce an 'antimuscarinic syndrome', characterized by psychosis and cognitive impairments. Latent inhibition (LI) is the poorer conditioning to a stimulus resulting from its non-reinforced pre-exposure. LI indexes the ability to ignore irrelevant stimuli and aberrations of this capacity produced by pro-psychotic agents (e.g. amphetamine, MK-801) are used extensively to model attentional impairments in schizophrenia. We recently showed that LI was disrupted by scopolamine at low doses, and this was reversed by typical and atypical antipsychotic drugs (APDs) and the acetylcholinesterase inhibitor physostigmine. Here, at a higher dose (1.5 mg/kg), scopolamine produced an opposite pole of attentional impairment, namely, attentional perseveration, whereby scopolamine-treated rats persisted in expressing LI under strong conditioning that prevented LI expression in controls. Scopolamine-induced persistent LI was reversed by cholinergic and glycinergic cognitive enhancers (physostigmine and glycine) but was resistant to both typical and atypical APDs (haloperidol and clozapine). The latter sets scopolamine-induced persistent LI apart from scopolamine- and amphetamine-induced disrupted LI, which are reversed by both typical and atypical APDs, as well as from other cases of abnormally persistent LI including MK-801-induced persistent LI, which is reversed by atypical APDs. Thus, scopolamine-induced persistent LI may provide a pharmacological LI model for screening cognitive enhancers that are efficient for the treatment of APD-resistant cognitive impairments in schizophrenia.

8.Bioequivalence and pharmacokinetic comparison of a single 200-mg dose of meclofenoxate hydrochloride capsule and tablet formulations in healthy Chinese adult male volunteers: a randomized sequence, open-label, two-period crossover study

Clinical Therapeutics. 30 (9) pp. 1651-7; 2008.

Jian-Jun Zou, Hong-Jian Ji, Ding-Wei Wu, Jing Yao, Qin Hu, Da-Wei Xiao and Guang-Ji Wang
Department of Clinical Pharmacology, Nanjing First Hospital of Nanjing Medical University, Nanjing, China.

Abstract
BACKGROUND: Meclofenoxate hydrochloride is a psychostimulant in the nootropic agent group available in capsule and tablet formulations approved for traumatic cataphora, alcoholic poisoning, anoxia neonatorum, and children's enuresis in China. Although these 2 generic formulations are marketed in China, information regarding their pharmacokinetics and bioequivalence in humans has not been published. OBJECTIVE: The aim of this study was to compare the pharmacokinetic properties and bioequivalence of the capsule (test) and tablet (reference) formulations of meclofenoxate hydrochloride 200 mg in healthy Chinese volunteers.

METHODS: This single-dose, randomized-sequence, open-label, 2-period crossover study was performed at the Nanjing First Hospital of Nanjing Medical University, Nanjing, China. Eligible subjects were healthy male volunteers who were randomly assigned at a 1:1 ratio to receive a single 200-mg dose of the test or reference formulation, followed by a 1-week washout period and administration of the alternate formulation. The study drugs were administered after a 12-hour overnight fast. As a prodrug, meclofenoxate is hydrolyzed into 4-chlorophenoxyacetic acid and is not detected in plasma. The active metabolite of meclofenoxate, chlorophenoxyacetic acid, was assayed using a high-performance liquid chromatography method. For analysis of pharmacokinetic properties, including Cmax, AUC0-24, and AUC0-infinity, blood samples were obtained at 0.33, 0.67, 1, 1.5, 2, 2.5, 3, 4, 6, 8, 10, 14, and 24 hours after administration. The formulations were considered bioequivalent if the log-transformed ratios of Cmax and AUC were within the predetermined equivalence range (80%-125%) as established by the US Food and Drug Administration (FDA). Subjects were interviewed concerning the occurrence of adverse events including excitement, insomnia, lassitude, and headache. Tolerability was assessed at baseline (before administration) and at 1, 2, 6, and 12 hours after administration by monitoring vital signs and laboratory tests (hematology, blood biochemistry, hepatic function, and urinalysis). RESULTS: Twenty-four Chinese male subjects (mean [range]age,23.5[22-30]years;weight,63.3[56-68]kg; height, 171 [165-184] cm) were enrolled; all completed the study. No period or sequence effect was observed. The 90% CIs for the log-transformed ratios of chlorophenoxyacetic acid Cmax, AUC0-24, and AUC0-infinity were 95.7 to 122.9, 97.6 to 111.9, and 97.8 to 111.7, respectively (all, P>0.05). Similar results were found for the data without log-transformation. No adverse events were reported or observed during this single-dose study. CONCLUSIONS: In this small study in healthy Chinese adult male volunteers, a single 200-mg dose of the capsule formulation was found to be bioequivalent to a single 200-mg dose of the tablet formulation based on the US FDA's regulatory definition (rate and extent of absorption). Both formulations were well tolerated.

9.Strain-dependent effects of cognitive enhancers in the mouse

Amino Acids. 34 (3) pp. 485-95; 2008.

B. Sunyer, S. Patil, C. Frischer, H. Hoeger and G. Lubec
Department of Pediatrics, Division of Pediatric Neuroscience, Medical University of Vienna, Vienna, Austria.

Abstract

A series of cognitive enhancers (CEs) have been reported to increase spatial memory in rodents, information on behavioral effects, however, is limited. The aim of the study was therefore to examine the behavioral effects of three CEs in two well-documented inbred mouse strains. C57BL/6J and DBA/2 mice were administered intraperitonial. D-cycloserine (DCS; NMDA receptor agonist), 1-(4-Amino-5-chloro-2-methoxyphenyl)-3-[1-butyl-4-piperidinyl]-1-propanone hydrochloride (RS67333; 5HT4-receptor agonist), and (R)-4-{[2-(1-methyl-2-pyrrolidinyl)ethyl]thio}phenol hydrochloride (SIB-1553A; beta-4-nicotinic receptor agonist) and tested in the open field (OF), elevated plus maze (EPM), neurological observational battery and rota-rod. Cognitive performance was tested in the Morris water maze. All compounds modified behavioral performance in the OF, DCS showed an anxiolytic effect in the EPM, and differences in the observational battery were observed i.e. vestibular drop was decreased by SIB-1553A and RS67333 treatment in C57BL/6J and increased with DCS treatment in DBA/2 mice. In the rota rod SIB-1553A improved motor performance.

DCS effects on learning and memory was comparable to controls whereas the other compounds impaired performance in the Morris water maze. In conclusion, behavioral testing of CEs in the mouse revealed significant changes that may have to be taken into account for evaluation of CEs, interpretation of cognitive studies and warrant further neurotoxicological studies. Moreover, strain-dependent differences were observed that in turn may confound results obtained from behavioral and cognitive testing.

10.The use of cognitive enhancers in behavioral disturbances of Alzheimer's disease

Consultant Pharmacist. 22 (9) pp. 754-62; 2007.

Lisa J. Miller
Michael E. DeBakey Veterans Affairs Medical Center, Houston, Texas 77030, USA. lisaj.miller@med.va.gov

Abstract
OBJECTIVE: To review the literature for double-blind, placebo-controlled trials that examined the efficacy of cognitive enhancers in the psychopathology of Alzheimer's disease. DATA SOURCES: Literature searches were conducted using MEDLINE and EMBASE databases and clinicaltrials.gov. STUDY SELECTION: Overall, 55 articles were reviewed for inclusion. Several open-label studies and case reports were found on this topic, but only those involving both tacrine and use of the Neuropsychiatric Inventory were included. Regarding other drugs, only double-blind, placebo-controlled trials were selected for inclusion. DATA SYNTHESIS: Limited data suggest that the anticholinesterase inhibitors and memantine offer an alternative or adjunct to the antipsychotics for the treatment of moderate-to-severe behaviors. The author reviewed the literature for pharmacological management of behavioral and psychological symptoms of dementia (BPSD) using these cognitive enhancers. CONCLUSION: The majority of patients with Alzheimer's disease will experience behavioral disturbances during the course of their disease. Atypical antipsychotics are used routinely in these situations to treat the psychotic features and agitation. However, atypicals now carry a "black box" warning issued by the Food and Drug Administration on the basis of evidence that their use in geriatric patients with dementia-related psychosis may put patients at increased risk of mortality as a result of cardiovascular or infectious events. An alternative to the atypicals may be the acetylcholinesterase inhibitors and memantine, which have been shown to stabilize cognitive as well as behavioral issues in patients, utilizing the "gold standard" for behavior, the Neuropsychiatric Inventory. Efficacy varies among agents, with the greatest positive effects seen with donepezil, which also has the greatest number of studies. Drug benefits were harder to demonstrate for mild-to-moderate BPSD compared with moderate-to-severe symptoms. [References: 48]

11.Piracetam--an old drug with novel properties?

Acta Poloniae Pharmaceutica. 62 (5) pp. 405-9; 2005.

Katarzyna Winnicka, Marian Tomasiak and Anna Bielawska
Department of Drug Technology, Medical University of Bialystok, 1 Kilinskiego Str., 15-089 Bialystok, Poland.

Abstract
Piracetam (2-oxo-1-pyrrolidine-acetamide), the most common of the nootropic drugs, is a cyclic derivative of gamma-aminobutyric acid. The treatment with piracetam improves learning, memory, brain metabolism, and capacity. Piracetam has been shown to alter the physical properties of the plasma membrane by increasing its fluidity and by protecting the cell against hypoxia. It increases red cell deformability and normalizes

aggregation of hyperactive platelets. Piracetam is an agent with antithrombotic, neuroprotective and rheological properties. The interaction of this molecule with the membrane phospholipids restores membrane fluidity and could explain the efficacy of piracetam in various disorders ranging from dementia and vertigo to myoclonus and stroke. [References: 40]

12.Inefficacy of piracetam in the prevention of painful crises in children and adolescents with sickle cell disease

Acta Haematologica. 113 (4) pp. 228-33; 2005.

R. C. Alvim, M. B. Viana, M. A. S. Pires, H. M. O. H. Franklin, M. J. Paula, A. C. Brito, T. F. Oliveira and P. V. Rezende
Department of Paediatrics and Haematology Division of Hospital das Clinicas, Federal University of Minas Gerais, Belo Horizonte, Brazil.

Abstract
Analgesia and hydration remain the only safe treatment for painful crises of sickle cell disease; hydroxyurea is effective, but the toxicity is still a problem. Piracetam is a nootropic drug that has reportedly been effective and non-toxic in sickle cell patients, but most studies were not placebo-controlled and included a small number of patients. The present study evaluated the drug in a double-blind crossed placebo-controlled clinical trial in 73 children and adolescents suffering from moderate to severe painful crises for 13 months. Information regarding frequency and severity of pain was acquired through monthly clinical evaluation, visits and house calls, and 4,300 weekly questionnaires filled out by the patients in their domiciles. A monthly pain score was calculated for each patient. Pain was the most frequent adverse manifestation of the disease stressing its significant bio-psycho-social impact. Although nearly all patients and relatives reported a better clinical course throughout the whole study, the drug was ineffective in the prevention of painful crises. This placebo effect may be ascribed to an unplanned and unsystematic 'cognitive-behavioural' management of the children. The pain score in the second semester of the study - both in the experimental and in the control groups - was significantly smaller than that in the first semester. In conclusion, piracetam was found to be ineffective in the prevention of painful crises; a powerful placebo effect due to adequate patient care was demonstrated. Copyright (c) 2005 S. Karger AG, Basel

13.Piracetam and TRH analogues antagonise inhibition by barbiturates, diazepam, melatonin and galanin of human erythrocyte D-glucose transport

British Journal of Pharmacology. 142 (3) pp. 594-608; 2004.

Richard J. Naftalin, Philip Cunningham and Iram Afzal-Ahmed
Physiology Division, Centre for Vascular Biology and Medicine, King's College London, Guy's Campus, New Hunt's House, London SE1 1UL. Richard.Naftalin@kcl.ac.uk

Abstract
1 Nootropic drugs increase glucose uptake into anaesthetised brain and into Alzheimer's diseased brain. Thyrotropin-releasing hormone, TRH, which has a chemical structure similar to nootropics increases cerebellar uptake of glucose in murine rolling ataxia. This paper shows that nootropic drugs like piracetam (2-oxo 1 pyrrolidine acetamide) and levetiracetam and neuropeptides like TRH antagonise the inhibition of glucose

transport by barbiturates, diazepam, melatonin and endogenous neuropeptide galanin in human erythrocytes in vitro. 2 The potencies of nootropic drugs in opposing scopolamine-induced memory loss correlate with their potencies in antagonising pentobarbital inhibition of erythrocyte glucose transport in vitro (P

17.A pilot trial of piracetam and ginkgo biloba for the treatment of cocaine dependence

Addictive Behaviors. 28 (3) pp. 437-48; 2003.

Kyle Kampman, Maria Dorota Majewska, Karen Tourian, Charles Dackis, James Cornish, Sabrina Poole and Charles O'Brien

Department of Psychiatry, University of Pennsylvania School of Medicine, Philadelphia, PA 19104, USA. kampman_k@mail.trc.upenn.edu

Abstract

BACKGROUND: Chronic cocaine use is associated with cognitive deficits that may reduce the effectiveness of psychosocial treatment and promote relapse in newly abstinent cocaine-dependent patients. Nootropic agents, such as piracetam and ginkgo biloba, may improve cognitive function and reduce the incidence of relapse in these patients. METHODS: This was a 10-week, double-blind, placebo-controlled pilot trial involving 44 cocaine-dependent subjects. Subjects received either piracetam (4.8 g/day), ginkgo biloba (120 mg/day), or placebo. Subjects were required to attain abstinence from cocaine during a 2-week baseline phase demonstrated by providing at least one benzoylecgonine (BE)-negative urine toxicology screen. Outcome measures included treatment retention, urine toxicology screens, Clinical Global Impression (CGI) scores, and results from the Addiction Severity Index (ASI). RESULTS: Ginkgo biloba was not superior to placebo in any outcome measure. Piracetam was associated with more cocaine use and lower CGI scores compared to placebo. CONCLUSIONS: Neither piracetam nor ginkgo biloba appears to be a promising medication for the treatment of cocaine dependence.

18.Smart drugs: tyrosine kinase inhibitors in cancer therapy

Cancer Cell. 1 (2) pp. 117-23; 2002.

Laura K. Shawver, Dennis Slamon and Axel Ullrich

SUGEN, Inc., 230 East Grand Avenue, South San Francisco, CA 94080, USA.

Abstract

Cancer therapy directed at specific, frequently occurring molecular alterations in signaling pathways of cancer cells has been validated through the clinical development and regulatory approval of agents such as Herceptin for the treatment of advanced breast cancer and Gleevec for chronic myelogenous leukemia and gastrointestinal stromal tumors. While most novel, target-directed cancer drugs have pregenomic origins, one can anticipate a postgenomic wave of sophisticated "smart drugs" to fundamentally change the treatment of all cancers. With these prospects, interest in this new class of therapeutics extends from basic research scientists to practicing oncologists and their patients. An extension of the initial successes in molecular oncology will occur more quickly and successfully through an appreciation of lessons learned with the first group of agents in their progress through clinical development. [References: 61]

19.Piracetam for dementia or cognitive impairment

Cochrane Database of Systematic Reviews. (2) pp. CD001011; 2001.

L. Flicker and G. Grimley Evans
Royal Perth Hospital, University of Western Australia, Box X2213 GPO, Perth, WA, Australia, 6847.
leonflic@rph.health.wa.gov.au

Abstract

OBJECTIVES: To determine the clinical efficacy of piracetam for the features of dementia or cognitive impairment, classified according to the major subtypes of dementia: vascular, Alzheimer's disease or mixed vascular and Alzheimer's disease, or unclassified dementia, or cognitive impairment not fulfilling the criteria for dementia. SEARCH STRATEGY: The trials were identified from a search of the Specialized Register of the Cochrane Dementia and Cognitive Improvement Group on 10 November 2000 using the term spiracetam, nootropic and 2-Oxo-1-pyrrolidine. In addition the pharmaceutical company responsible for marketing most of the piracetam worldwide, UCB Pharma, provided a comprehensive list of abstracts, which included many unpublished studies. As many of these unpublished, placebo-controlled studies will be reviewed as possible. SELECTION CRITERIA: All unconfounded trials specified as randomized in which treatment with piracetam was administered for more than a day and compared with placebo in patients with dementia of Alzheimer type, vascular dementia,or mixed vascular and Alzheimer's disease, or unclassified dementia, or cognitive impairment not fulfilling the criteria for dementia. DATA COLLECTION AND ANALYSIS: Data were extracted independently by two reviewers. Each study was independently verified as fulfilling the inclusion criteria. Studies were rated for methodological quality by assessment of blinding and loss before analysis as described by Jadad et al. (1996). Studies were pooled if appropriate and possible, and the pooled odds ratios (95%CI) or the average differences (95%CI) were estimated. Where possible, intention-to-treat analyses were undertaken. Sensitivity analyses were performed to determine if successive elimination of those studies performing most poorly on these quality criteria changed the effect estimate. MAIN RESULTS: Unfortunately, many of the studies were of cross-over design and first-phase data were unavailable, or could not be extracted. Global Impression of Change was the only outcoeme for which there was a significant volume of evidence from the pooled data. There was evidence of heterogeneity in the results from the individual studies, chi-square test = 20.8 (df=5). Using a fixed effects model the odds ratio for improvement in the piracetam group compared with the placebo group was 3.55, [95% CI][2.45, 5.16]. If a random effects model was used the odds ratio was 3.47 [1.29, 9.30]. If one single-blind study was excluded, the fixed effects model yielded an odds ratio of 3.36 [2.29, 4.99] and if a random effects model was applied then the odds ratio was 2.89 [1.01, 8.24]. The evidence of effects on cognition and other measures, was inconclusive. REVIEWER'S CONCLUSIONS: At this stage the evidence available from the published literature does not support the use of piracetam in the treatment of people with dementia or cognitive impairment. Although effects were found on global impression of change, no benefit was shown by any of the more specific measures. There is a need for further evaluation of piracetam by : 1) Obtaining the data from the identified studies for an individual patient database review, 2) Performing a randomized trial of piracetam in patients with diagnoses made by currently accepted diagnostic criteria. The trial should extend over for a period of at least 6 months and preferably longer. Specific cognitive instruments which are sensitive to change, Clinician Global Impression of Change,

levels of dependency and caregiver quality of life scales should also be incorporated in such a study. [References: 89]

20.Piracetam affects facilitatory I-wave interaction in the human motor cortex

Clinical Neurophysiology. 112 (2) pp. 275-9; 2001.

S. Wischer, W. Paulus, M. Sommer and F. Tergau
Department of Clinical Neurophysiology, University of Goettingen, Robert-Koch-Strasse 40, D-37075, Goettingen, Germany.

Abstract
OBJECTIVE: To investigate by means of transcranial magnetic stimulation (TMS) the effect of a single oral dose of the GABA derivate piracetam on intracortical facilitatory I-wave interaction. METHODS: The study was performed in 8 healthy volunteers. Before, 1, 3, 6, and 24 h after intake of 4000 mg piracetam, MEPs in the relaxed abductor digiti minimi muscle were elicited by a recently introduced double pulse TMS technique with a suprathreshold first and a subthreshold second stimulus. From interstimulus intervals of 0.5-5.1 ms 3 periods were observed in which MEP facilitation showed maxima - so-called peaks of I-wave interaction - and which were separated by two troughs with no facilitation. We studied the changes in timing and size of the peaks over time. RESULTS: With piracetam, I-wave peaks showed a reduction in size as well as a shortening of the latencies at which the peaks occurred. Both changes were significant at 6 h after drug intake compared to baseline. The effects were partially reversible after 24 h. CONCLUSIONS: The mode of action of piracetam within the nervous system is almost unknown. The peak size reduction was similar to effects that were seen under GABAergic drugs, although GABAergic properties of piracetam have not been observed so far. Shortening of the I-wave peak latencies is a new phenomenon. The results are discussed on the basis of the known therapeutic effects of piracetam in cortical myoclonus and as nootropic agent.

21.Piracetam and levetiracetam: close structural similarities but different pharmacological and clinical profiles

Epileptic Disorders. 2 (2) pp. 99-105; 2000.

P. Genton and B. Van Vleymen
Centre Saint-Paul, Marseille, France. piergen@aol.com

Abstract
Piracetam (PIR) and levetiracetam (LEV), an S-enantiomer, are pyrrolidone derivatives that share similar chemical structures but have distinct pharmacological profiles and consequently different clinical uses. Although the mode of action of neither drug has been fully elucidated, they do not interact with inhibitory or excitatory neurotransmission or alter membrane excitability. A brain-specific stereoselective binding site has been identified for which LEV and other S-enantiomers, but not PIR, have high affinity. In preclinical studies, PIR significantly improves learning and memory; in contrast, LEV has less effect but is much more active in preventing seizures. Both drugs have a high therapeutic index and are well tolerated. PIR, a nootropic drug, is used in the therapy of age-related cognitive disturbances and poststroke aphasia. Clinical experience has also shown that at high doses it is effective against cortical myoclonus. LEV is an antiepileptic drug. Clinical trials have confirmed its efficacy in partial seizures and preliminary findings suggest that it is also effective in generalized seizures and myoclonus. [References: 47]

22.Piracetam in the treatment of acute stroke

Pharmacopsychiatry. 32 Suppl 1 pp. 25-32; 1999.

J. M. Orgogozo
Department of Neurology, Pellegrin Hospital, University of Bordeaux II, France.

Abstract
The neuroprotective properties of the nootropic agent piracetam together with reported hemorrheologic and antithrombotic effects provided the rationale for the evaluation of piracetam in acute stroke. Pilot studies showed an increase in compromised regional cerebral blood flow and improvement in motor function, aphasia and level of consciousness. Subsequently the Piracetam in Acute Stroke Study (PASS) was performed and the chief results have recently been reported (Stroke 28 (1997) 2347-2352). This was a multicenter double-blind trial in 927 patients to determine whether, compared with placebo, piracetam improved outcome when given within 12 hours of the onset of acute ischemic stroke, confirmed by computed tomography within 24 hours of admission (but not necessarily prior to treatment). Patients received an initial iv bolus of placebo or 12g piracetam, 12g piracetam daily for 4 weeks and maintenance treatment for a further 8 weeks. Neurologic status at 4 weeks was the primary end point; secondary outcome measures were functional outcome and aphasia at 12 weeks. Results in aphasic patients have not previously been reported. Analysis was planned both in all patients (n = 927) and an early treatment subgroup (n = 460) treated within 6 hours of stroke onset. This period was subsequently redefined as 7 hours. Intention-to-treat analyses in the total population showed a significant (P = 0.04) increase compared with placebo in the number of patients recovered from aphasia but no significant neurologic or functional improvement. Post hoc analysis in the early treatment subgroup showed improved neurologic outcome (P = 0.07), better function (P = 0.02) and a greater recovery rate from aphasia (P = 0.02). Additional analysis in this early treatment subgroup confined to 360 patients with moderate and severe stroke showed significant improvement in all 3 outcomes. There was no significant difference in mortality between treatment groups after 12 weeks. There were fewer deaths in piracetam-treated patients in those patients in the intention-to-treat population admitted with primary hemorrhagic stroke.

23.[The 1998 domestic state of development of cognitive enhancers]

Nippon Yakurigaku Zasshi - Folia Pharmacologica Japonica. 114 (6) pp. 327-36; 1999.

M. Hasegawa, Y. Noda, Y. Maeda, K. Yamada and T. Nabeshima
Department of Neuropsychopharmacology, Nagoya University Graduate School of Medicine, Japan.

Abstract
Recently, there is a great social problem that geriatric disorders, especially senile dementia, are growing rapidly with the increasing percentage of aged individuals in the total population. However, there is yet no drug that has reliable effects on senile dementia on the market. Therefore, society requires the development of new drugs that can support patients so that they can smoothly live their daily lives by themselves. In this study, we attempted to investigate the 1998 domestic state of development of cognitive enhancers by summarizing many publications and by gathering questionnaires from pharmaceutical, food, synthetic fiber and chemical manufacturing companies. There were 40 currently investigated cognitive enhancers in Japan as of the end of March, 1999 including 37 newly synthesized compounds and 3 new dosage forms or applications

of additional effects. On the classification according to the mechanism of drug action, 12 of the investigated drugs are cholinomimetic agents, 12 are ameliorators of neuronal transmission, 1 is an intracellular mediative substance, 3 are neuropeptides, 2 are cerebral metabolic activators, 2 are cerebral circulation enhancers and 7 are neuronal cell protectors, and 1 is another type. For dementia of the Alzheimer's type, there are 1, 3, 10 and 2 drugs in prerecognition, phase late II, early II and I of clinical trials, respectively. For cerebrovascular dementia and cerebrovascular disease, there are 19 drugs being investigated. Seven of these compounds such as E2020 (donepedil HCl), DM-9384 (nefiracetam), TA-0910 (taltirelin), NS-3 (montirelin), TTC-909 (clinprost), DR-3305 (ebselen) and AVS (nicaraven) are at the prerecognition stage for marketing. It is important that effective cognitive enhancers will be supplied for clinical stage use as soon as possible.

24.Piracetam in the treatment of schizophrenia: implications for the glutamate hypothesis of schizophrenia
Journal of Clinical Pharmacy & Therapeutics. 24 (5) pp. 369-74; 1999.
A. A. Noorbala, S. Akhondzadeh, R. Davari-Ashtiani and H. Amini-Nooshabadi
Roozbeh Psychiatric Hospital, Tehran University of Medical Sciences, Tehran, Iran.
Abstract
OBJECTIVE: There is a growing interest in investigating the role of glutamate receptors in the pathophysiology of schizophrenia. Indeed, the hyperdopaminergic theory of schizophrenia can explain only the positive symptoms of schizophrenia, whereas the glutamate hypothesis may provide a more comprehensive view of the illness. We undertook a trial to investigate whether the combination of haloperidol with piracetam, a nootropic agent which modulates the glutamate receptor positively was more effective than haloperidol alone. METHODS: Thirty patients who met the DSM IV criteria for schizophrenia completed the study. Patients were allocated in a random fashion, 14 to haloperidol 30 mg/day plus piracetam 3200 mg/day and 16 to haloperidol 30 mg/day plus placebo. RESULTS: Although both protocols significantly decreased the score of the positive symptoms, the negative symptoms, the general psychopathological symptoms and the total score of PANSS scale over the trial period, the combination of haloperidol and piracetam showed a significant superiority over haloperidol alone in the treatment of schizophrenic patients. CONCLUSION: Piracetam, a member of the nootropic class of drugs and a positive modulator of glutamate receptor, may be of therapeutic benefit in treating schizophrenic patients in combination with typical neuroleptics. However, a larger study to confirm our results is warranted

25.Single-dose piracetam effects on global complexity measures of human spontaneous multichannel EEG
International Journal of Psychophysiology. 34 (1) pp. 81-7; 1999.
I. Kondakor, C. M. Michel, J. Wackermann, T. Koenig, H. Tanaka, J. Peuvot and D. Lehmann
The KEY Institute for Brain-Mind Research, University Hospital of Psychiatry, Zurich, Switzerland.
Abstract
Global complexity of 47-channel resting electroencephalogram (EEG) of healthy young volunteers was studied after intake of a single dose of a nootropic drug (piracetam, Nootropil UCB Pharma) in 12 healthy volunteers. Four treatment levels were used: 2.4, 4.8, 9.6 g piracetam and placebo. Brain electric activity was assessed through Global Dimensional Complexity and Global Omega-Complexity as quantitative measures of the complexity of the trajectory of multichannel EEG in state space. After oral ingestion (1-1.5 h), both measures

showed significant decreases from placebo to 2.4 g piracetam. In addition, Global Dimensional Complexity showed a significant return to placebo values at 9.6 g piracetam. The results indicate that a single dose of piracetam dose-dependently affects the spontaneous EEG in normal volunteers, showing effects at the lowest treatment level. The decreased EEG complexity is interpreted as increased cooperativity of brain functional processes.

26.['Smart drugs' enticements on the Internet]
Nederlands Tijdschrift voor Geneeskunde. 142 (17) pp. 977-80; 1998.
J. M. Keppel Hesselink
Abstract
'Smart drugs' are drugs supposed to enhance certain physical or mental functions, without harmful side effects. The target group for the misleading advertisement campaigns on the Net is mostly the younger generation. Smart drugs clearly are pharmacologically active: phenytoin, selegilin and growth hormone are examples of compounds that can be ordered from mail order pharmacies without prescription. Misleading medical information on the Internet is undesirable and potentially harmful and the widespread availability of smart drugs should be realised by pharmacists and physicians. Screening the Net for this kind of information should be implemented as a standard procedure and specific, targeted information to put these drugs in a realistic perspective could be the first step leading to 'debunking' the Internet.

27.Functional activation of cerebral blood flow abolished by scopolamine is reversed by cognitive enhancers associated with cholinesterase inhibition: a positron emission tomography study in unanesthetized monkeys
Journal of Pharmacology & Experimental Therapeutics. 281 (3) pp. 1408-14; 1997.
H. Tsukada, T. Kakiuchi, I. Ando and Y. Ouchi
Central Research Laboratory, Hamamatsu Photonics K.K., Shizuoka, Japan.
Abstract
The effects of somatosensory stimulation on the regional cerebral blood flow (rCBF) response were studied in unanesthetized monkeys before and after treatment with scopolamine and three cognitive enhancers (physostigmine, E2020 and tacrine) that inhibit cholinesterase, using 15O-labeled water and high-resolution positron emission tomography. Under control conditions, somatosensory stimulation induced a significant increase in the rCBF response in the contralateral somatosensory cortex of monkey brain. Intravenous administration of scopolamine (50 microg/kg) resulted in abolishment of the rCBF response to stimulation. The rCBF response abolished by pretreatment with scopolamine was recovered by administration of physostigmine (1 or 10 microg/kg), E2020 (10 or 100 microg/kg) or tacrine (100 or 1000 microg/kg), in a dose-dependent manner. The effect of E2020 (100 microg/kg) on the rCBF response lasted for >4 hr, whereas the effects of physostigmine and tacrine were of shorter duration. These findings suggest that these compounds reversed the scopolamine-abolished rCBF response to somatosensory stimulation via enhancement of cholinergic neurotransmission, which was mainly induced by cholinesterase inhibition.

28.[A critical review of available tools for evaluating memory enhancers in Alzheimer's disease]
Revue de Medecine Interne. 18 (1) pp. 59-71; 1997.

J. Poitrenaud, M. Kalafat, L. Israel and D. Guez
Service des explorations fonctionnelles, hopital Charles-Foix, Ivry-sur-Seine, France.

Abstract
The study reviews critically the neuropsychological measurement tools available to assess the efficacy of potential memory enhancers in patients with Alzheimer's disease (AD). A review of some concepts concerning the characteristics of memory impairment in AD patients and of problems related to the selection of a study population provides a framework for this analysis. The tools available for efficacy evaluations are then classified according to their conceptual and formal characteristics: psychometric tests, rating scales and clinician global assessment tools. In each category the instruments were assessed according to several criteria: construct and ecological validity; sensitivity to changes in symptomatology; inter-rater reliability and the reliability of equivalent forms; feasibility for use with AD patients, especially brevity, to encourage their cooperation. Examining advantages, drawbacks and limitations of instruments in each category yields two conclusions: no single instrument scale or test battery fills all the criteria for a clinical trial, but these imperfect instruments are complementary. Accordingly, a battery including the three categories of instruments and allowing for investigation of the different memory systems has been proposed. [References: 59]

29.Alterations in acetylcholine, NMDA, benzodiazepine receptors and protein kinase C in the brain of the senescence-accelerated mouse: an animal model useful for studies on cognitive enhancers

Behavioural Brain Research. 83 (1-2) pp. 51-5; 1997.

Y. Nomura, Y. Kitamura, T. Ohnuki, T. Arima, Y. Yamanaka, K. Sasaki and Y. Oomura
Department of Pharmacology, Faculty of Pharmaceutical Sciences, Hokkaido University, Sapporo, Japan.

Abstract
The senescence-accelerated mouse (SAMP8) is a useful murine model of accelerated aging and learning deficiency. We examined bindings of [3H]pirenzepine, [3H]dizocilpine (MK-801), [3H]flunitrazepam, [3H]8-hydroxy-2-(di-n-propylamino)-tetralin (8-OH-DPAT) and [3H]phorbol 12,13-dibutylate (PDBu) in SAMP8 brains, and compared them to those of SAMR1 (control). In the hippocampus of SAMP8 at 12 months, bindings of [3H]pirenzepine, [3H]MK-801, [3H]flunitrazepam, [3H]8-OH-DPAT and [3H]PDBu were significantly lower than those in SAMR1. In the cerebral cortex, bindings of [3H]pirenzepine, [3H]flunitrazepam and [3H]8-OH-DPAT were higher in SAMP8 than in SAMR1 at 12 months. [3H]PDBu binding was decreased in both the fractions of the membrane and cytosol in the hippocampus of SAMP8. The neurochemical findings presented here support behavioral and pharmacological findings that SAMP8 is a useful model of learning dysfunction and anxiety-deficiency. The usefulness of SAMP8 in studies on cognitive enhancers is also discussed.

30.Cognitive enhancers and hippocampal long-term potentiation in vitro

Behavioural Brain Research. 83 (1-2) pp. 45-9; 1997.

S. Kaneko, T. Maeda and M. Satoh
Department of Pharmacology, Faculty of Pharmaceutical Sciences, Kyoto University, Japan.

Abstract
We review our works on the pharmacological modulation of long-term potentiation (LTP) at guinea pig hippocampal mossy fiber-CA3 synapses in vitro. The magnitude of tetanus-induced LTP at the mossy fiber

synapse was augmented by perfusion of slices with several cognitive enhancers, such as bifemelane (1 microM). The mossy fiber LTP was enhanced by somatostatin (0.32 microM) and inhibited in somatostatin-depleted slices from cysteamine-treated guinea pigs. An involvement of the 5-HT3 receptor also showed that granisetron (0.1 microM) enhanced the mossy fiber LTP. The above-mentioned enhancements by perfused agents were commonly reversed, at least in part, by muscarinic antagonists. However, the magnitude of mossy fiber LTP was bidirectionally modulated by muscarinic stimulations of slices with physostigmine or carbachol at different concentrations. The enhancing effects of high-concentration carbachol was antagonized by pirenzepine, and in contrast, the inhibition by low-concentration carbachol was antagonized in the presence of AF-DX116. When guinea pigs were preinjected with the cholinotoxin AF64A, the magnitude of LTP was decreased in the slices prepared from AF64A-treated animals. These results suggest that endogenous acetylcholine dominantly plays facilitatory roles through muscarinic M1 receptors in the induction of mossy fiber LTP. The pharmacological characterization of mossy fiber LTP may be of help to the evaluation of cognitive enhancers at a neuronal circuit level. [References: 33]

31.Cognitive enhancers in theory and practice: studies of the cholinergic hypothesis of cognitive deficits in Alzheimer's disease

Behavioural Brain Research. 83 (1-2) pp. 15-23; 1997.

T. W. Robbins, G. McAlonan, J. L. Muir and B. J. Everitt

Department of Experimental Psychology, University of Cambridge, UK. TWR2@cus.cam.ac.uk

Abstract

The current status of the cholinergic hypothesis of cognitive dysfunction in Alzheimer's disease is reviewed in the context of recent attempts to alleviate specific cognitive impairments produced in rats by excitotoxic lesions of basal forebrain neurons by treatment with cholinergic agents. AMPA-induced lesions of the nucleus basalis region in rats produce profound and relatively specific reductions in neocortical markers of cholinergic function but fail to affect performance in many tests of memory and learning in rats. However, such lesions produce specific deficits in responding accurately in a test of visual attentional performance, which are reversed dose-dependently by treatment with systemic physostigmine or nicotine. Analogous improvements have been reported in a clinical trial of the anticholinesterase tacrine in patients with Alzheimer's disease. By contrast, AMPA-induced lesions of the medial septum produce profound reductions in hippocampal acetylcholine and accompanying delay-dependent deficits in a delayed non-matching-to-position procedure which measures spatial working memory in rats. This impairment is shown to be reversed to some extent by treatment with low doses of physostigmine. The results are discussed in terms of the multivariate nature of the neurochemical pathology of Alzheimer's disease and attendant limitations in the use of the cholinergic strategy. The cognitive costs, as well as benefits, of cognitive enhancers are discussed, as well as the need to broaden our therapeutic approach to other neurotransmitter systems and other neurodegenerative disorders. [References: 34]

32.Nootropic agents stimulate neurogenesis

Expert Opinion on Therapeutic Patents. 19 (5) pp. 727-730; 2009.

P. Taupin

(Taupin) Dublin City University, School of Biotechnology, Glasnevin, Dublin 9, Ireland P. Taupin, Dublin City University, School of Biotechnology, Glasnevin, Dublin 9, Ireland. E-mail: philippe.taupin@dcu.ie

Abstract

The application is in the field of adult neurogenesis, neural stem cells and cellular therapy. It aims to characterize the activity of nootropic agents on adult neurogenesis in vitro. Nootropic agents are substances improving cognitive and mental abilities. AMPA (alpha-amino-3-hydroxyl-5-methyl-4- isoxazolepropionate) and nootropic agents were assessed for the potential to differentiate human neural progenitor and stem cells into neuronal cells in vitro. They were also tested for their behavioural activity on the novel object recognition task. AMPA, piracetam, FK-960 and SGS-111 induce and stimulate neuronal differentiation of human-derived neural progenitor and stem cells. SGS-111 increases the number of visits to the novel object. The neurogenic activity of piracetam and SGS-111 is mediated through AMPA receptor. The neurogenic activity of SGS-111 may contribute and play a role in its nootropic activity. These results suggest that nootropic agents may elicit some of their effects through their neurogenic activity. The application claims the use of nootropic agents for their neurogenic activity and for the treatment of neurological diseases, disorders and injuries, by stimulating or increasing the generation of neuronal cells in the adult brain. 2009 Informa UK Ltd. All rights reserved.

33.Epibatidine - Design for new cognitive enhancers. [Czech] Epibatidin - Model pro nova kognitiva

Psychiatrie. 5 (SUPPL. 2) pp. 99-100; 2001.

J. Patocka, J. Fusek and M. V. Vazquez

(Patocka, Fusek, Vazquez) Katedra Toxikologie, Vojenska Lekarska Akademie, 500 01 Hradec Kralove, Czech Republic J. Patocka, Katedra Toxikologie, Vojenska Lekarska Akademie, 500 01 Hradec Kralove, Czech Republic. E-mail: patocka@pmfhk.cz

Abstract

Epibatidine is an animal alkaloid, found in poisonous secretion of Ecuadorian poison-dart frog Epipedobates tricolor. Its chemical structure is similar to nicotine as well as its pharmacological properties. Epibatidine is a selective agonist of nicotinic acetylcholine receptors with high affinity to some their subtype. Degenerative disorders with loss of memory are characterized by low density of nicotinic receptors in brain and compounds with stimulation effect on these receptors are treat new and hopeful group of cognitive enhancers. Epibatidine is useful designated model for the development of this therapeutics.

34.Surfing with side effects. Problems with smart drugs. [Dutch] Surfen Met Bijwerkingen. Problemen Rond Smart Drugs

Pharmaceutisch Weekblad. 133 (43) pp. 1598-1604; 1998.

J. M. Keppel

(Keppel) Medisch Bioloog en Arts-farmocoloog (Keppel) Bayer AG Werkt (Keppel) Universiteit van Witten/Herdecke, Germany J.M. Keppel, Laboratorium Pharmakologie, Universitat Witten/Herdecke, Stockemerstrasse 10, 158448 Witten, Germany

Abstract

Smart drugs are drugs that are supposed to be of use to enhance certain physical or mental functions, without harmful side effects. The target group for the misleading advertisement campaigns on the internet are mostly

the younger generation. Smart drugs clearly are pharmacologically active compounds that can be ordered via mail order pharmacies without prescription. Misleading medical information on the internet is undesirable and potentially harmful and the wide-spread availability of smart drugs should be realised by pharmacists. In this article the advertisements for phenytoin, selegiline and gammahydroxybutyrate are discussed as examples of claimed effects and potential hazards.

35.Piracetam: A review of its clinical potential in the management of patients with stroke

CNS Drugs. 9 (6) pp. 497-511; 1998.

S. Noble and P. Benfield

(Noble, Benfield) Adis International Limited, Auckland, New Zealand (Noble) Adis International Limited, 41 Centorian Drive, Mairangi Bay, Auckland 10, New Zealand S. Noble, Adis International Limited, 41 Centorian Drive, Mairangi Bay, Auckland 10, New Zealand. E-mail: demail@adis.co.nz

Abstract

Piracetam is a derivative of -aminobutyric acid (GABA) and was the first commercially available nootropic drug. Although widely evaluated in the treatment of senile cognitive disorders and dyslexia, it has also been assessed as a treatment for deficits associated with acute stroke. Data from a subgroup of patients in the Piracetam in Acute Stroke Study (PASS) suggest that piracetam may provide a modest level of neurological and functional protection when administered within several hours of a moderate or severe stroke. However, findings from this study have yet to be confirmed. Data from a number of small, short term studies in patients treated within a few days of stroke suggest that piracetam is more effective than placebo for the treatment of functional deficits. Piracetam has shown varying degrees of efficacy against poststroke aphasia in several placebo-controlled studies when administered within several hours, a few days or several weeks or more after stroke. Its most likely role is one of early treatment (within several hours or a few days of stroke) in combination with speech therapy. Conclusions: On the basis of current information, it seems reasonable to conclude that piracetam may have potential in the treatment of general poststroke deficits, but that further data (including those from an ongoing phase III study) are needed before more definitive conclusions can be drawn. The available data need to be considered against the absence of established treatment options in this indication and the generally modest efficacy of other investigational agents. The benign tolerability profile of piracetam is also relevant in this respect. Piracetam may be a useful option in patients with poststroke aphasia, especially when started soon after stroke, particularly since there are no established pharmacological treatment options for this indication and speech therapy has only limited success.

36.Piracetam in the treatment of acute stroke

CNS Drugs. 9 (SUPPL. 1) pp. 41-49; 1998.

J. M. Orgogozo

(Orgogozo) Department of Neurology, Pellegrin Hospital, University of Bordeaux II, Bordeaux, France J.-M. Orgogozo, Department of Neurology, Pellegrin Hospital, University of Bordeaux II, Bordeaux, France

Abstract

The documented neuroprotective properties of the nootropic agent piracetam together with reported haemorrheological and antithrombotic effects provided the rationale for the evaluation of piracetam in acute

stroke. Pilot studies showed an increase in compromised regional cerebral blood flow and improvement in functional ability, aphasia and level of consciousness with piracetam. Subsequently, the Piracetam in Acute Stroke Study (PASS) was performed and the principal results have recently been reported. The aim of this multicentre double-blind trial in 927 patients was to determine whether, compared with placebo, piracetam improved outcome when given within 12 hours of the onset of acute ischaemic stroke, confirmed by computed tomography within 24 hours of admission (not necessarily prior to treatment). Patients received an initial intravenous bolus of placebo or piracetam 12g, piracetam 12g daily for 4 weeks and maintenance treatment (4.8g daily) for a further 8 weeks. Neurological status at 4 weeks was the primary end-point; secondary outcome measures were functional outcome and aphasia at 12 weeks. Results in patients with aphasia have previously not been reported. Analysis was planned in all patients (n = 927) and in an early treatment subgroup (n = 460), which was treated within 6 hours of stroke onset, a period subsequently redefined as 7 hours. Intention-to-treat analyses in the total population showed a significant ($p > 0.04$) increase, compared with placebo, in the number of patients who recovered from aphasia, but no significant change in neurological or functional outcome. Post hoc analysis in the early treatment subgroup showed improved neurological outcome ($p = 0.07$), better functional ability ($p = 0.02$) and a greater recovery rate from aphasia ($p = 0.02$). Additional analysis confined to 360 patients with moderate and severe stroke in this early treatment subgroup showed significant improvement in all outcomes. There was no significant difference in mortality between treatment groups after 12 weeks. There were fewer deaths in piracetam-treated patients in the intention-to-treat population admitted with primary haemorrhagic stroke. The PASS confirmed the improvement in aphasia seen in pilot studies with piracetam, and showed an improvement in neurological outcome and better functional ability with piracetam in patients treated within 7 hours of stroke onset, particularly those with moderate and severe stroke. A randomised doubleblind trial that aims to prospectively confirm these findings is in progress.

37.Pharmacological properties of piracetam: Rationale for use in stroke patients

CNS Drugs. 9 (SUPPL. 1) pp. 19-27; 1998.

G. Hitzenberger

(Hitzenberger) Society of Clinical Pharmacology, Vienna, Austria G. Hitzenberger, Gesellschaff fur Klin. Pharmakologie, Kinderspitalgasse 10/15-17, A-1090 Wien, Austria

Abstract

Piracetam, the standard nootropic drug, is known to improve a whole series of mental activities, particularly higher cortical functions. Recently, this drug has come under investigation as a cerebroprotective agent in ischaemic stroke. In vitro studies showed cerebral metabolism-stimulating effects, which formed the basis for the interpretation of in vivo studies. These studies have demonstrated some benefits under experimental conditions of brain-lesioning influences (e.g. central ischaemia, hypoxia, etc.). Piracetam has possible neuromodulatory, cerebrovascular and electrophysiological effects, as well as beneficial therapeutic effects on the microcirculation. Its haemorrheological and antithrombotic properties suggest the possible use of piracetam in ischaemic cerebral infarction and as a cerebroprotective agent in stroke. Published data have shown that the pharmacokinetic properties of piracetam in animals and humans are similar after oral and intravenous administration, resulting in almost complete oral bioavailability.

38.'Smart drugs': Misleading advertising on internet. [Dutch] 'Smart drugs': De verlokkingen van het internet

Nederlands Tijdschrift voor Geneeskunde. 142 (17) pp. 977-980; 1998.

J. M. K. Hesselink

(Hesselink) Griffensteynselaan 68, 3703 AD Zeist, Netherlands J.M.K. Hesselink, Arts-farmacoloog, Griffensteynselaan 68, 3703 AD Zeist, Netherlands

Abstract

'Smart drugs' are drugs supposed to enhance certain physical or mental functions, without harmful side effects. The target group for the misleading advertisement campaigns on the Net is mostly the younger generation. Smart drugs clearly are pharmacologically active: phenytoin, selegilin and growth hormone are examples of compounds that can be ordered from mail order pharmacies without prescription. Misleading medical information on the Internet is undesirable and potentially harmful and the widespread availability of smart drugs should be realised by pharmacists and physicians. Screening the Net for this kind of information should be implemented as a standard procedure and specific, targeted information to put these drugs in a realistic perspective could be the first step leading to 'debunking' the Internet.

39.Treatment of acute ischemic stroke with piracetam

Stroke. 28 (12) pp. 2347-2352; 1997.

P. P. De Deyn, J. De Reuck, W. Deberdt, R. Vlietinck, J. M. Orgogozo and B. Raoult

(De Reuck) Department of Neurology, University Hospital, Ghent, Belgium (Deberdt) Dept. of Clin. R. and D., UCB Pharma, Braine-l'Alleud, Belgium (Orgogozo) Department of Neurology, Pellegrin Hospital, University of Bordeaux II, Bordeaux, France (De Deyn) Department of Neurology, University of Antwerp, Universiteitsplein 1, 2610 Wilrijk, Antwerp, Belgium (Raoult) UCB Pharma, Building S3, Chemin du Foriest, 1420 Braine-l'Alleud, Belgium P.P. De Deyn, Department of Neurology, University of Antwerp, Universiteitsplein 1, 2610 Wilrijk, Antwerp, Belgium

Abstract

Background and Purpose: Piracetam, a nootropic agent with neuroprotective properties, has been reported in pilot studies to increase compromised regional cerebral blood flow in patients with acute stroke and, given soon after onset, to improve clinical outcome. We performed a multicenter, randomized, double-blind trial to test whether piracetam conferred benefit when given within 12 hours of the onset of acute ischemic stroke to a large group of patients.' Methods: Patients received placebo or 12 g piracetam as an initial intravenous bolus, 12 g daily for 4 weeks and 4.8 g daily for 8 weeks. The primary end point was neurologic outcome after 4 weeks as assessed by the Orgogozo scale. Functional status at 12 weeks as measured by the Barthel Index was the major secondary outcome. CT scan was performed within 24 hours of the onset of stroke but not necessarily before treatment. Analyses based on the intention to treat were performed in all randomized patients (n=927) and in an 'early treatment' population specified in the protocol as treatment within 6 hours of the onset of stroke but subsequently redefined as less than 7 hours after onset (n=452). Results: In the total population, outcome was similar with both treatments (the mean Orgogozo scale after 4 weeks: piracetam 57.7, placebo 57.6; the mean Barthel Index after 12 weeks: piracetam 55.8, placebo 53.1). Mortality at 12 weeks was 23.9% (111/464) in the piracetam group and 19.2% (89/463) in the placebo group (relative risk 1.24, 95% confidence interval, 0.97 to 1.59; P=.15). Deaths were fewer in the piracetam group in those

patients in the intention- to-treat population admitted with primary hemorrhagic stroke. Post hoc analyses in the early treatment subgroup showed differences favoring piracetam relative to placebo in mean Orgogozo scale scores after 4 weeks (piracetam 60.4, placebo 54.9; P=.07) and Barthel Index scores at 12 weeks (piracetam 58.6, placebo 49.4; P=.02). Additional analyses within this subgroup, confined to 360 patients with moderate and severe stroke (initial Orgogozo scale score

40.A critical review of available measurement tools for assessing memory enhancers in patients with Alzheimer's disease. [French] Une revue critique des instruments disponibles pour l'evaluation des medicaments promnesiants dans la maladie d'Alzheimer

Revue de Medecine Interne. 18 (1) pp. 59-71; 1997.

J. Poitrenaud, M. Kalafat, L. Israel and D. Guez

(Poitrenaud) Serv. des Explorations F., Hopital Charles-Foix, 7, avenue de la Republique, 94205 Ivry-sur-Seine, France (Kalafat) Service de Neurologie 3, Hopital de la Salpetriere, 47-83, boulevard de l'Hopital, 75651 Paris Cedex 13, France (Israel) Institut de Psychologie, Universite Lumiere-Lyon 2, 5 avenue Pierre-Mendes-France, 69679 Bron Cedex, France (Guez) Inst. de Rech. Intl. Servier, 6, place des Pleiades, 92415 Courbevoie, France J. Poitrenaud, Serv. des Explorations Fonct., Hopital Charles-Foix, 7, Avenue de la Republique, 94205 Ivry-sur-Seine, France

Abstract

The study reviews critically the neuropsychological measurement tools available to assess the efficacy of potential memory enhancers in patients with Alzheimer's disease (AD). A review of some concepts concerning the characteristics of memory impairment in AD patients and of problems related to the selection of a study population provides a framework for this analysis. The tools available for efficacy evaluations are then classified according to their conceptual and formal characteristics: psychometric tests, rating scales and clinician global assessment tools. In each category the instruments were assessed according to several criteria: construct and ecological validity; sensitivity to changes in symptomatology; inter-rater reliability and the reliability of equivalent forms; feasability for use with AD patients, especially brevity, to encourage their cooperation. Examining advantages, drawbacks and limitations of instruments in each category yields two conclusions: no single instrument scale or test battery fills all the criteria for a clinical trial, but these imperfect instruments are complementary. Accordingly, a battery including the three categories of instruments and allowing for investigation of the different memory systems has been proposed.

41.Piracetam in patients with chronic vertigo. Results of a double-blind, placebo-controlled study

Clinical Drug Investigation. 11 (5) pp. 251-260; 1996.

U. Rosenhall, W. Deberdt, U. Friberg, A. Kerr and W. Oosterveld

(Rosenhall, Deberdt, Friberg, Kerr, Oosterveld) UCB Pharma, International Development, Chemin du Foriest, B-1420 Braine-l'Alleud, Belgium W. Deberdt, UCB Pharma, International Development, Chemin du Foriest, B-1420 Braine-l'Alleud, Belgium

Abstract

The nootropic agent piracetam, which exerts diverse effects through actions on cerebral neurotransmission, has been reported to alleviate vertigo. We performed a multicentre, double-blind, placebo-controlled study to

assess the efficacy and tolerability of piracetam 800 mg 3 times daily orally for 8 weeks. The study group consisted of 143 middle-aged and elderly outpatients of ear, nose and throat clinics who had suffered from vertigo for at least 3 months, had experienced at least 3 episodes per month, and the vertigo was severe enough to disrupt daily life. Primary outcome measures were patient self-evaluations of vertigo: the frequency of episodes, and their severity using visual analogue scales (VAS). Malaise and imbalance between episodes (VAS), the effect of vertigo on walking (VAS), the duration of incapacity, and overall evaluations by patients and investigators were also assessed. On entry, episodes were more frequent ($p < 0.05$) and malaise between episodes more severe ($p < 0.05$) in the piracetam group. Data were not evaluable in 54 patients because of either adverse events (12 piracetam, 12 placebo) or protocol deviations. An intention-to-treat analysis showed that episodes of vertigo were less frequent ($p < 0.03$) but not less severe on piracetam than on placebo: interval malaise ($p < 0.05$) and imbalance ($p < 0.01$) improved more and the duration of incapacity was less ($p < 0.05$). These changes, which were maximal after 8 weeks' medication, had almost disappeared 4 weeks after the end of treatment. Tolerance to piracetam was good, with few drug-related adverse events occurring. These findings provide further evidence that piracetam alleviates vertigo by reducing the frequency of episodes, the severity of malaise and imbalance between episodes, and the duration of associated incapacity.

42.Cortical myoclonus: A valid human model for the demonstration of the neuromodulatory effects of piracetam

Acta Therapeutica. 21 (1) pp. 77-84; 1995.

D. Karacostas, N. Artemis and S. Giannopoulos

(Karacostas, Artemis, Giannopoulos) B' Dept. of Neurology, AHEPA Hospital, Aristotelian Univ. School of Med., Thessaloniki, Greece D. Karacostas, B' Dept. of Neurology, AHEPA Hospital, Aristotelian Univ. School of Med., Thessaloniki, Greece

Abstract

Piracetam, a gamma aminobutyric acid (GABA) analogue and the prototype 'nootropic' drug, has been widely used in the treatment of memory and cognitive deficits in several disease entities including Alzheimer's disease. Since 1978, case reports and clinical trials have shown the therapeutic efficacy of piracetam in patients with myoclonus of diverse aetiology, especially of cortical origin. Experimentally, piracetam seems to increase brain energy metabolism, hippocampal acetylcholine release and the firing of neurons in the locus coeruleus. Its effects are either additive to or antagonise the actions of glutamate, GABA and acetylcholine and are directed towards specific membrane sites. Published data support the view that cortical myoclonus represents a valid human model for demonstrating the modulatory effects of piracetam on neurotransmission.

43.Piracetam in developmental reading disorders: A review

European Child and Adolescent Psychiatry. 3 (2) pp. 59-71; 1994.

C. R. Wilsher and E. A. Taylor

(Wilsher, Taylor) Department of Psychology, University College of North Wales, 69 Clarendon Road, Watford, Hertfordshire WD1 1DJ, United Kingdom C.R. Wilsher, Department of Psychology, University College of North Wales, 69 Clarendon Road, Watford, Hertfordshire WD1 1DJ, United Kingdom

Abstract
This paper reviews the studies that have examined piracetam, the first of the nootropic drugs, as a treatment for developmental reading disorder. In various studies, 751 children have completed treatment in controlled double-blind trials using standardised tests of reading as outcome measures. Quantitative and qualitative review suggests that the findings are best interpreted as showing a statistical superiority of piracetam over placebo. Mechanism of action and clinical significance are discussed. The effect size is modest, but the drug is well tolerated and the balance of benefit over hazard is encouraging for future use.

44.Effect of piracetam on recovery and rehabilitation after stroke: A double- blind, placebo-controlled study
Clinical Neuropharmacology. 17 (4) pp. 320-331; 1994.
P. Enderby, J. Broeckx, W. Hospers, F. Schildermans and W. Deberdt
(Enderby, Broeckx, Hospers, Schildermans, Deberdt) International Development, UCB Pharma, Chemin du Foriest, B-1420 Braine-l'Alleud, Belgium W. Deberdt, International Development, UCB Pharma, Chemin du Foriest, B-1420 Braine-l'Alleud, Belgium
Abstract
The nootropic agent piracetam has been shown to improve learning and memory, and it may, by this means, facilitate recovery and rehabilitation after a stroke. We report the results of a pilot study exploring its effects in patients undergoing rehabilitation after acute cerebral infarction in the carotid artery territory. We compared piracetam and placebo, each given for 12 weeks, in a multicenter, double-blind, randomized trial of parallel-group design; testing was performed at baseline (6-9 weeks poststroke), weeks 5 and 12, and, in fewer patients, 12 weeks after termination of treatment. Standardized tests of activities of daily living (Barthel Index, Kuriansky Test), aphasia (Aachen Aphasia Test), and perception (Rivermead Perception Assessment Battery) were the primary efficacy variables. Of 158 patients, 137 (81 males, 56 females) were studied after treatment and 88 at 24-week follow- up. Thirty patients on piracetam (45%) and 37 on placebo (53%) were aphasic on entry. Both groups, including the subgroups with aphasia, were well matched at baseline for demographic data, stroke sequelae, type and severity of aphasia, and prognostic parameters. Multivariate analysis of Aachen Aphasia subtest scores showed a significant overall improvement relative to baseline in favor of piracetam ($p = 0.02$) at 12 weeks. This was not seen at 24 weeks when, however, fewer patients were available for evaluation so that we could neither confirm nor deny whether improvement was maintained after cessation of piracetam. We were unable to demonstrate an effect on tests of activities of daily living and could neither confirm nor exclude an effect on perceptual deficit. We have shown an improvement in aphasia in patients undergoing rehabilitation after a stroke after 12 weeks' treatment with piracetam that requires confirmation in further studies.

45.Superiority of Antagonic-Stress composition versus nicergoline in gerontopsychiatry
Annals of the New York Academy of Sciences. 717 pp. 332-342; 1994.
F. Schneider, R. Popa, G. Mihalas, P. Stefaniga, I. G. Mihalas, R. Maties and R. Mateescu
(Schneider, Popa, Mihalas, Stefaniga, Mihalas, Maties, Mateescu) Department of Physiology, University of Medicine and Pharmacy, Eftimie Murgu Square no 2, RO-1900 Timisoara, Romania F. Schneider, Department of Physiology, University of Medicine and Pharmacy, Eftimie Murgu Square no 2, RO-1900 Timisoara, Romania

Abstract
Nicergoline (NE) - a cerebral vasodilator with nicotinic acid esterified in its molecule - and Antagonic-Stress (AS) composition - a neurometabolic nootropic, also containing nicotinic acid but with fast and prolonged release - were evaluated in senile dementia of Alzheimer's type (SDAT), mild to moderate intensity (DSM-IV Options Book, 1991 and ICD-10, 1990 criteria). A double-blind, randomized, comparative, and parallel clinical trial was performed on 62 old people divided into 2 groups and exclusively treated with NE or AS. Psychogeriatric evaluations (Sandoz Clinical Assessment-Geriatric scale, Self-Assessment Scale-Geriatric and their subscales) and psychometric tests (digit symbol of WAIS, Wechsler Memory Scale, and Wechsler Adult Intelligence Scale - WAIS) were made before and after 3 months of treatment. Prolonged and large dose treatments with NE and AS significantly decreased the psychogeriatric scores, diminished the deterioration index, and improved cognitive performances (ANOVA). Therapeutical effects of AS were significantly higher than those of NE (ANCOVA). The better actions of AS in senile dementia and for improving cognitive function and behavior are discussed in connection with its multiple neurometabolic composition, the synergism of components, the antiischemic action of its antioxidants, its anti-free radical complementary action (deceleration of the aging rate, brain and erythrocyte lipofuscinolysis, complex antioxidative and scavenger formula), the multivitamin and multimineral supplementation and, finally, with the superiority of multitherapy vs. monotherapy.

46.Smart Drugs: Implications of student use

Journal of Primary Prevention. 14 (3) pp. 197-207; 1994.

R. J. Canterbury and E. Lloyd
(Canterbury, Lloyd) BRH Division, Substance Abuse Studies Institute, University of Virginia, Charlottesville, VA 22901, United States E. Lloyd, BRH Division, Substance Abuse Studies Institute, University of Virginia, Charlottesville, VA 22901, United States

Abstract
The use of Smart Drugs to enhance intelligence, improve memory and maximize cognitive functioning in healthy individuals has attracted the attention of the popular press. This paper discusses the implication of the nonmedical college student use of 'nootropic' Smart Drugs, a class of pharmaceuticals legally available in other countries to treat diseases associated with mental decline or dysfunction. Nootropic drug use is compared to steroid use in a student population. In a survey of college students, 5% of the males reported casual use of a drug to increase their intelligence, enhance their memory and make them smarter: 2.5% of these students probably used a nootropic drug.

47.Effects of oxiracetam on organic brain syndrome following activation of the high-affinity choline uptake system: A neurochemical analysis of the cerebrospinal fluid of four patients

Current Therapeutic Research - Clinical and Experimental. 54 (2) pp. 194-201; 1993.

A. Borromei, R. Caramelli, M. Cerisoli, R. Gaggi, A. M. Gianni, A. Lozito, B. Vargiu and C. Alvisi
(Borromei, Caramelli, Cerisoli, Gaggi, Gianni, Lozito, Vargiu, Alvisi) Istituto di Clinica Neurologica, Via Ugo Foscolo 7, 40123 Bologna, Italy A. Borromei, Istituto di Clinica Neurologica, Via Ugo Foscolo 7, 40123 Bologna, Italy

Abstract
Nootropic drugs stimulate acetylcholine (ACh) synthesis by activating the high-affinity choline uptake system. Cerebrospinal fluid (CSF) samples were obtained from four patients with organic brain syndrome in order to gauge acetylcholinesterase activity and measure levels of ACh, 5- hydroxyindoleacetic acid, vanillylmandelic acid, and 5-hydroxytryptophan before and at the end of nootropic treatment. Patients were given oxiracetam for 30 days at dosages varying from 1600 mg/day to 2000 mg/day. Clinical evaluations conducted at the beginning and end of treatment were made by measuring the Sandoz Clinical Assessment-Geriatric (SCAG) score and reaction times to simple auditory and visual stimuli. Following oxiracetam treatment, average ACh levels in CSF decreased significantly, and there was a concomitant improvement in the clinical evaluation (SCAG).

48.Long-term and high-dose piracetam treatment of Alzheimer's disease
Neurology. 43 (2) pp. 301-305; 1993.
B. Croisile, M. Trillet, J. Fondarai, B. Laurent, F. Mauguiere and M. Billardon
(Croisile, Trillet, Fondarai, Laurent, Mauguiere, Billardon) Department of Neurology, Hopital Neurologique, 59 boulevard Pinel, 69003 Lyon, France B. Croisile, Department of Neurology, Hopital Neurologique, 59 boulevard Pinel, 69003 Lyon, France
Abstract
Preclinical research suggests that piracetam (a nootropic drug) may improve cognitive functions, but previous studies have failed to demonstrate a clear benefit for the treatment of Alzheimer's disease (AD). We report a 1- year, double-blind, placebo-controlled, parallel-group study with a high dose of piracetam (8 g/d per os) in 33 ambulant patients with early probable AD. Thirty subjects completed the 1-year study. No improvement occurred in either group, but our results support the hypothesis that long-term administration of high doses of piracetam might slow the progression of cognitive deterioration in patients with AD. The most significant differences concerned the recall of pictures series and recent incident and remote memory. The drug was well-tolerated.

49.Effect of CDP-choline on learning and memory processes in rodents
Methods and Findings in Experimental and Clinical Pharmacology. 14 (8) pp. 593-605; 1992.
V. D. Petkov, A. H. Mosharrof, R. Kehayov, V. V. Petkov, E. Konstantinova and D. Getova
(Petkov, Mosharrof, Kehayov, Petkov, Konstantinova, Getova) Institute of Physiology, Bulgarian Academy of Sciences, Sofia, Bulgaria V.D. Petkov, Institute of Physiology, Bulgarian Academy of Sciences, Sofia, Bulgaria
Abstract
The effects of cytidine (5') diphosphocholine (CDP-choline) on learning and memory were studied using conditioned reflex methods for passive avoidance and active avoidance with punishment reinforcement (step-through, step-down, shuttle box and maze), for active avoidance with alimentary reinforcement (staircase maze), and the Morris water maze. The majority of experiments involved comparative studies of the nootropic drugs meclofenoxate and/or piracetam. CDP-choline was administered orally, in some of the experiments also intraperitoneally, at doses of 10-500 mg/kg body weight once or twice daily for 5 or 7 days. In separate cases only single doses were administered. Trainings started one hour after the last dose of the drugs. Retention

tests were given 3 h, 24 h, 7 days or 10 days after training. The results obtained with the different methods document CDP-choline's ability to improve learning and memory in rats and mice. No essential differences in the effects of CDP-choline were established upon oral and intraperitoneal administration of the drug. The learning- and memory-facilitating effects of CDP-choline were similar to those of meclofenoxate and piracetam. The results of the present study permit us to define CDP-choline as a substance capable of improving cognitive levels.

50.Clinical use of piracetam in epileptic patients

Current Therapeutic Research - Clinical and Experimental. 52 (3) pp. 355-360; 1992.

H. R. Chaudhry, N. Najam, C. De Mahieu, A. Raza and N. Ahmad
(Chaudhry, Najam, De Mahieu, Raza, Ahmad) Department of Psychiatry, Fatima Jinnah Medical College, 83-Shah Jamal Colony, Lahore 54000, Pakistan H.R. Chaudhry, Department of Psychiatry, Fatima Jinnah Medical College, 83-Shah Jamal Colony, Lahore 54000, Pakistan

Abstract
In a double-blind, randomized, controlled study, we investigated the effects of piracetam, a nootropic compound, on cognitive disturbances and the potency of antiepileptic drugs in 60 epileptic patients and 15 age-matched control subjects. Each subject was given the following tests at the time of entry and 3 and 5 weeks after the start of treatment: memory for design test, Wechsler Intelligence Scale for Children/Wechsler Adult Intelligence Scale subtests, electroencephalography, and a neurologic examination. Our findings suggest that piracetam has a positive influence on cognitive measures in epileptic patients. In addition, piracetam does not diminish the antiepileptic effects of anticonvulsant drugs. Instead, it improved seizure protection in patients receiving carbamazepine.

51.Effects of four non-cholinergic cognitive enhancers in comparison with tacrine and galanthamine on scopolamine-induced amnesia in rats

Psychopharmacology. 106 (1) pp. 26-30; 1992.

P. Chopin and M. Briley
(Chopin, Briley) Centre Recherche Pierre Fabre, 17 Avenue Jean Moulin, F-81106 Castres, Netherlands M. Briley, Centre Recherche Pierre Fabre, 17 Avenue Jean Moulin, F-81106 Castres, Netherlands

Abstract
Amnesia can be induced in rats in the passive avoidance paradigm by administration of scopolamine, a central muscarinic receptor antagonist. Tacrine or galanthamine, inhibitors of acetylcholinesterase, given in conjunction with scopolamine partially reversed the scopolamine-induced deficit in passive avoidance performance. Four so-called cognitive enhancers, all widely used for the treatment of the symptoms associated with mental aging, cerebral insufficiency and senile memory disorder, were investigated in this paradigm. Piracetam, an extract of Ginkgo biloba, dihydroergocristine and a combination of raubasine with dihydroergocristine, all attenuated the amnesia induced by scopolamine. In contrast, nicergoline had no significant effect. Raubasine alone also failed to significantly attenuate scopolamine-induced amnesia, although some doses of raubasine had a non-significant tendency ($P < 0.10$) to reduce the amnesia.

52.Piracetam. An overview of its pharmacological properties and a review of its therapeutic use in senile cognitive disorders

Drugs & aging. 1 (1) pp. 17-35; 1991.

M. W. Vernon and E. M. Sorkin

(Vernon, Sorkin) Adis International Drug Information Services, Auckland, New Zealand. M.W. Vernon, Adis International Drug Information Services, Auckland, New Zealand.

Abstract

Piracetam is the first of the so-called 'nootropic' drugs, a unique class of drugs which affect mental function. In animal models and in healthy volunteers, the drug improves the efficiency of the higher telencephalic functions of the brain involved in cognitive processes such as learning and memory. The pharmacology of piracetam is unusual because it protects against various physical and chemical insults applied to the brain. It facilitates learning and memory in healthy animals and in animals whose brain function has been compromised, and it enhances interhemispheric transfer of information via callosal transmission. At the same time, even in relatively high dosages it is devoid of any sedative, analeptic or autonomic activities. How piracetam exerts its effects on memory disorders is still under investigation, although among other proposed mechanisms of action it is thought to facilitate central nervous system efficiency of cholinergic neurotransmission. Results from trials involving elderly patients with senile cognitive disorders have been equivocal, as have the results obtained when piracetam has been combined with acetylcholine precursors. Piracetam seems to be almost completely devoid of adverse effects, and is extremely well tolerated. In conclusion, opinion is divided as to the benefits of piracetam in the treatment of senile cognitive decline. Although double-blind studies in the elderly have produced mixed results, some such trials (particularly those involving larger numbers of patients) have reported favourable findings, thus offering some reason for cautious optimism in a notoriously difficult area of therapeutics. However, further investigations of piracetam alone and in combination therapy are required before any absolute conclusions can be drawn.

53.Clinical issues of cognitive enhancers in Alzheimer disease

Alzheimer Disease and Associated Disorders. 5 (1 SUPPL.) pp. S25-S31; 1991.

M. Calvani and A. Carta

(Calvani, Carta) Neurological Research Department, Sigma Tau S.p.A., Via Pontina km. 30,400, 00040 Pomezia, Rome Italy Neurological Research Department, Sigma Tau S.p.A., Via Pontina km. 30,400, 00040 Pomezia, Rome Italy

Abstract

Cognitive enhancers is a provocative and vague label for drugs used to treat dementia of the Alzheimer type. Several issues have to be carefully considered in order to perform reliable clinical trials with such compounds. The lack of an animal model appropriately matching the human pathology, the difficulty in finding worldwide criteria for clinical diagnosis and determining which patients are eligible and how they are best tested, methods of treatment, and interpreting results are undoubtedly the major problems to be solved. A review of the literature points out that the ''day-after'' approach of treatment (once severe neuropathological damage has been established) is no longer feasible, or has limited advantages. A different pharmacological approach, based on preventive measures during the first stages of the neurodegeneration, seems mandatory.

54.The use of a scopolamine model to study the potential nootropic effects of aniracetam and piracetam in healthy volunteers

Journal of Psychopharmacology. 4 (4) pp. 219-232; 1990.

K. Wesnes, R. Anand, P. Simpson and L. Christmas

(Wesnes, Anand, Simpson, Christmas) Cognitive Drug Research, 13 The Grove, Reading RG1 4RB United Kingdom Cognitive Drug Research, 13 The Grove, Reading RG1 4RB United Kingdom

Abstract

In this study 26 healthy volunteers received scopolamine 0.7 mg subcutaneously on seven occasions at least a week apart. Cognitive efficiency was measured with a test battery before and 60 min following scopolamine on each occasion. Following this, over the seven occasions, a range of oral and intravenous dose regimens were administered including aniracetam 2 mg intravenously, 100 mg intravenously, 200 mg intravenously, 1500 mg per os and piracetam 2400 mg per os. On each session the test battery was then performed again at 120 and 200 min following scopolamine. The seven treatments were administered double-blind and the order was counterbalanced between volunteers over visits using a Latin Square design. At 60 min, scopolamine produced marked and significant decrements in all of the measures of memory and information processing. Aniracetam 1500 mg was able to significantly antagonize decrements on both memory and information processing tasks. The other active treatments also produced significant effects, but for two these were equal to, and for two slightly above, the number which may have occurred by chance, and thus were questionable. Overall, the findings demonstrate that aniracetam 1500 mg can antagonize cognitive decrements produced by cholinergic blockade in healthy volunteers, and suggest that the drug possesses nootropic properties.

55.A controlled trial of piracetam in intellectually impaired patients with Parkinson's disease

Movement Disorders. 5 (3) pp. 230-234; 1990.

M. Sano, Y. Stern, K. Marder and R. Mayeux

(Sano, Stern, Marder, Mayeux) Department of Neurology, Columbia University, New York, NY United States Department of Neurology, Columbia University, New York, NY United States

Abstract

Twenty patients with Parkinson's disease and marked intellectual impairment or dementia participated in a double-blind placebo controlled trial of the nootropic, piracetam. A standardized neurological examination, a neuropsychological test battery, and a functional scale, The Sickness Impact Profile, were completed for all patients. They were then assigned by blind randomization to drug or placebo conditions receiving 3.2 g of piracetam or an identical amount of placebo for 12 weeks. The dose was increased to 4.8 g for an additional 12 weeks. Neurological, psychological, and functional measures were rated as improved, unchanged, or worsened in comparison to baseline performance. Twenty-five percent of the patients did not complete the trial for reasons unrelated to the medication. Although there was a significant improvement on one subtest of the functional scale, no significant effects were demonstrated in cognitive or neurological measures.

56.Choosing a target for cognitive enhancers

Clinical Therapeutics. 12 (4) pp. 298-305; 1990.

H. Allain, D. Bentue-Ferrer and R. Decombe

(Allain, Bentue-Ferrer, Decombe) Laboratory of Experimental and Clinical Pharmacology, Pontchaillou Hospital, Rennes France Laboratory of Experimental and Clinical Pharmacology, Pontchaillou Hospital, Rennes France

Abstract

The development of drugs to improve cognitive functions (cognitive enhancers) has been hampered by the failure to define pharmacologic targets. The justifications for the development of cognitive enhancers for medicine, the pharmaceutical industry, and society are outlined. The procedures required fro the development of these drugs and their therapeutic objectives are discussed. It is concluded that a better definition of the direct targets of cognitive enhancers can guarantee cost and time savings for the industry and produce a therapeutically effective agent.

57.An hypothesis on the role of glucose in the mechanism of action of cognitive enhancers

Psychopharmacology. 99 (4) pp. 431-438; 1989.

G. L. Wenk

(Wenk) Neuromnemonics Laboratory, Department of Psychology, Johns Hopkins University, Baltimore, MD 21218 United States Neuromnemonics Laboratory, Department of Psychology, Johns Hopkins University, Baltimore, MD 21218 United States

Abstract

This review presents evidence that some cognition enhancing drugs produce their beneficial effects on learning and memory by increasing the availability of glucose for uptake and utilization into the brain. The hypothesis further suggests that many cognition enhancing drugs act through a peripheral mechanism rather than directly on the brain. The general hypothesis is supported by four independent and converging pieces of evidence: 1) Some cognition enhancing drugs may not cross the blood-brain barrier, but can still facilitate memory; 2) Some cognition enhancing drugs are effective only when injected peripherally, but not when injected directly into the brain; 3) Many cognition enhancing drugs are not effective after adrenalectomy; 4) Cognitive function is correlated with glucose regulation in aged animals and humans. These four lines of research have implications for the role of glucose in the action of specific cognitive enhancers.

58.Piracetam: physiological disposition and mechanism of action

Advances in neurology. 43 pp. 675-685; 1986.

M. T. Tacconi and R. J. Wurtman

M.T. Tacconi

Abstract

Piracetam, the prototype of nootropic drugs, is claimed to have therapeutic value in some patients suffering from cognitive deficits, especially when hypoxia may be a factor in the deficits. The drug lacks sedative or stimulatory effects and apparently does not influence behavior in normal people. The site of action and the mechanism by which piracetam exerts its beneficial effects are not well established. This review has summarized experimental and clinical data on its absorption, distribution, and excretion, and described the biochemical data available that bear on its mechanism of action.

59.Piracetam combined with lecithin in the treatment of Alzheimer's disease

Neurobiology of Aging. 7 (4) pp. 269-276; 1986.

J. H. Growdon, S. Corkin, F. J. Huff and T. J. Rosen
(Growdon, Corkin, Huff, Rosen) Department of Neurology, Harvard Medical School at Massachusetts General Hospital, Boston, MA 02114 United States Department of Neurology, Harvard Medical School at Massachusetts General Hospital, Boston, MA 02114 United States

Abstract
Nootropics, a new class of drugs believed to activate mental functions, have been proposed as a treatment for clinical disorders in which cognition is impaired. We therefore administered the nootropic drug piracetam, alone and in combination with phosphatidylcholine (PC), to 18 patients with Alzheimer's disease (AD), and measured the effects of treatment on a broad range of cognitive functions. Piracetam was administered according to three double-blind crossover protocols and a replication study that differed in piracetam dose (2.4 to 9.9 g/day) and whether PC (18 g/day) was administered concurrently. The drug was well tolerated, and there were no toxic side effects. Plasma choline levels rose significantly during piracetam and PC administration; monoamine metabolites in cerebrospinal fluid were unaffected by treatment. Piracetam, either alone or in combination with PC, did not significantly affect cognition in the AD group as a whole, nor did it improve test performance in any single patient.

60.The red cell as a model for studying neurotropic drugs: Effects of piracetam on erythrocyte membrane

Acta Therapeutica. 9 (1) pp. 75-83; 1983.

M. Rubegni, G. Ravenni and D. Provvedi
(Rubegni, Ravenni, Provvedi) Geriat. Div., Reg. Gen. Hosp., Siena Italy Geriat. Div., Reg. Gen. Hosp., Siena Italy

Abstract
Red blood cells, like platelets, can be utilized as a pharmacological model in various disorders of the nervous system. In this study the activity of piracetam, a nootropic agent, on hemolysis induced by streptolysin and on red blood cell ageing in vitro has been studied. Protection from the hemolysis induced by streptolysin and a prolongation of red blood cell survival time have been shown. The slowing down of the acetylcholinesterase activity after piracetam, particularly evident when glycerol is added to the incubation medium, is evidence in favor of an activity of piracetam on the red blood cells membrane.

61.The effects of oxiracetam (ISF 2522) in patients with organic brain syndrome. (A double-blind controlled study with piracetam)

Drug Development Research. 2 (5) pp. 447-461; 1982.

T. M. Itil, G. N. Menon, M. Bozak and A. Songar
(Itil, Menon, Bozak, Songar) Div. Biol. Psychiatry, New York Med. Coll., New York, NY 10591 United States Div. Biol. Psychiatry, New York Med. Coll., New York, NY 10591 United States

Abstract
In a double-blind controlled trial, the clinical effects of oxiracetam, a new 'nootropic' compound, were investigated in a group of 60 elderly patients with organic mental disorders (DSM-11). The starting dose of both oxiracetam and the control drug, piracetam, was 400 mg. The dosage was increased by 400 mg at weekly

intervals up to 2,400 mg daily (sixth week). During the following 6 weeks the administered dose was fixed at 2,400 mg daily. Most of the important target symptoms improved significantly over time, both subjectively (i.e., rating scales) and objectively (i.e., psychological tests), after administration of either oxiracetam or piracetam. In comparison to piracetam, oxiracetam exhibited more statistically significant improvement in the memory factor, whereas piracetam showed more improvement than oxiracetam in factors of paranoid ideation and agitation. Both drugs were tolerable and did not elicit any significant side effects. It was postulated that 'nootropics' may represent a new group of CNS effective compounds, and thus be a 'second generation' of psychotropics, which have more direct effects on the central target organs than are presently found in the 'classical' psychotropics.